

Enterprise Home Lab Network Design Project

Documentation

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Purpose: Enterprise-style home lab for networking, systems administration, virtualization, monitoring, automation, and cloud engineering.

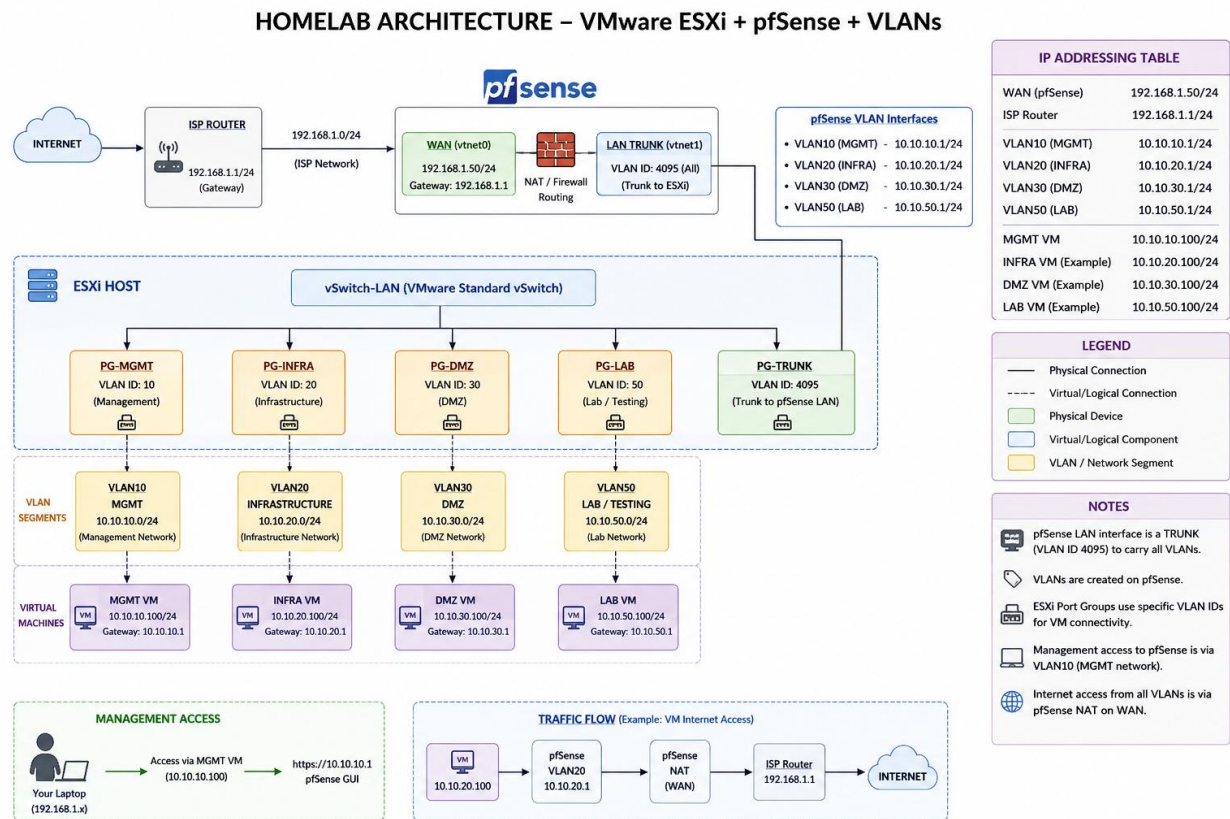
1. Executive Summary

This lab is designed as a professional enterprise environment running on VMware ESXi. Primary goals include CCNA/CCNP preparation, network automation, monitoring, cloud integration, security testing, and portfolio development.

2. Hardware Platform

Hypervisor	CPU	RAM	Storage	Physical NICs	Edge Device	Firewall
VMware ESXi 8.x	6 Cores	32 GB	319 GB SSD	2	ISP Router	pfSense VM

3. High Level Architecture



4. VLAN Design

Network segmentation is implemented to:

- Improve security
- Reduce broadcast domains
- Simulate enterprise architecture
- Enable firewall policy enforcement
- Support future cloud integration

ID	Caption	Network
VLAN10	MGMT	10.1.10.0/24
VLAN20	INFRA	10.1.20.0/24
VLAN30	DMZ	10.1.30.0/24
VLAN40	CLOUD	10.1.40.0/24
VLAN50	N-LAB	10.10.50.0/24
VLAN-TRUNK	pfSense LAN	10.1.1.1

VLAN Purpose

VLAN 10 – Management

Purpose: Administrative access to infrastructure.

Contains:

- Management Workstation
- VMware Management

VLAN 20 – Infrastructure

Purpose: Core Infrastructure Services.

Contains:

- Active Directory
- DNS
- DHCP
- Certificate Services
- File Services
- Network Monitoring Application

VLAN 30 – DMZ

Purpose: Internet facing applications.

Contains:

- Web Servers
- Test Applications
- Reverse Proxy

VLAN 40 – CLOUD

Purpose: Hybrid Cloud Integrations.

Contains:

- Linux
- Azure Connectors
- AWS Integration
- Cloud Monitoring Agents

VLAN 50 – Network Lab

Purpose: Network testing environment.

Contains:

- EVE-NG/GNS3, Routers, Switches, Network Topology, Network Configurations.

5. ESXi Design

Virtual Switch Architecture

vSwitch-WAN

Purpose: External connectivity.

Connected to: **vmnic0**

Port Groups

- PG-WAN

vSwitch-LAN

Purpose: Internal virtual Infrastructure.

Connected to: **no physical uplink**

Port Groups

- PG-TRUNK (4095)
- PG-MGMT (10)
- PG-INFRA (20)
- PG-DMZ (30)
- PG-CLOUD (40)
- PG-LAB (50)

NOTE: Internal routing occurs through pfSense

6. Firewall Design

pfSense

WAN Interface

Connected to: PG-WAN

IP Address: DHCP from ISP Router

LAN Interface (Act as a Trunk carrier)

Connected to: PG-TRUNK (VLAN 4095)

IP Address: Manually assigned

VLAN Interfaces

Interface	VLAN	Gateway
MGMT_VLAN10	10	10.1.10.0
INFRA_VLAN20	20	10.1.20.0
DMZ_VLAN30	30	10.1.30.0
CLOUD_VLAN40	40	10.1.40.0
NLAB_VLAN50	50	10.10.50.0

Firewall Policy

WAN

Allow:

- Established connections
- VPN Access

Block:

- All Unsolicited Traffic

Management

Allow:

- SSH
- HTTPS
- RDP

Infrastructure

Allow:

- Domain Services
- DNS
- Internal Communication
- Monitoring

DMZ

Allow:

- Internet Access
- Required Internal Access Only

N-LAB

Allow:

- Network Simulation
- Controlled Infrastructure Access
- Monitoring

7. IP Addressing Plan

Management

Network: 10.1.10.0/24

Gateway: 10.1.10.1

Reserved: 10.1.10.1-50

DHCP:

10.1.10.100-199

Infrastructure

10.1.20.0/24

Static Allocation Only

N-Lab

10.10.50.0/24

Dynamic Allocation Allowed

8. Security Architecture

- Zone-based segmentation using VLAN.
- Inter-VLAN routing through pfSense.
- Management access restricted to MGMT VLAN.
- Future WireGuard VPN, Suricata IDS/IPS, and PKI deployment.

9. Monitoring Architecture

TOOLS

SolarWinds

Monitoring:

- ESXi
- Windows Servers
- pfSense
- Cisco Devices
- Juniper Devices

LibreNMS

Monitoring:

- SNMP
- Syslog
- NetFlow

Grafana

Visualization:

- Dashboards
- Capacity Planning
- Performance Analytics

10. Automation Architecture

Platform

AUTO01

Services:

- Git
 - Ansible
 - Docker
 - Python
 - Terraform
-

Automation Use Cases

Network Automation

- VLAN Deployment
- Interface Configuration
- Configuration Backup

Server Automation

- User Creation
- Software Deployment
- Patch Management

Monitoring Automation

- Alert Creation
- Device Onboarding

11. Backup and Disaster Recovery

- Weekly configuration backups.
- Monthly VM exports.
- GitHub repository backup.
- Documented recovery procedures.

12. Implementation Roadmap

Phase 1 - Foundation

Phase 2 - Core Infrastructure

Phase 3 - Network Engineering Lab

Phase 4 - Monitoring

Phase 5 - Automation and DevOps

Phase 6 - Cloud Integration

13. Operations Standards

- Naming conventions
- Change management
- Documentation standards
- Configuration management
- Version control through GitHub

14. Success Criteria

The project is considered successful when:

- All VLANs are operational
- Inter-VLAN routing works
- Internet access is available
- Active Directory is deployed
- SolarWinds is monitoring infrastructure
- EVE-NG devices are monitored
- Git-based documentation is maintained
- AWS/Azure integrations are functional

15. Future Enhancements

- Kubernetes
- Security Onion
- Azure Integration
- AWS Integration
- RADIUS/TACACS+
- Advanced Monitoring